File Systems

Operating system provides a uniform logical view of information storages. OS abstracts from the physical properties of its storage device to define a logical storage unit which is called as file.

Files are mapped by the operating system onto physical devices. This storage devices are usually nonvolatile, so the contents are persistent through power failure and system reboots.

A file is a named collection of related information that is recorded on secondary storage.

File Attributes :

A file has certain attributes which may vary from one operating system to another.

(1) Name : The symbolic file name is the only information kept in human readable format.

(2) Identifier : This is a unique number which identifies the file within the file system.

(3) Type : This identifies the type of file.

(4) Location : This information is a pointer to a device and location of the file on that device.

(5) Size : The current size of the file.

(6) Protection : Access control information determines who can do reading, writing, executing so on.

(7) Time date and user identification :

This information is kept for creation, last modification and last use.

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File Operation :

A file is an abstract datatype. To define a file properly, we need to consider the operation that can be performed on files.

(1) **creating a file :** It is a two step process
   i. find the space for file
   ii. make entry in the directory.

(2) **Writing a file :** A system call is used for writing.

(3) **Reading a file :** A system call is used with a read pointer.

(4) **Repositioning within a file :**
This operation reposition the current file position to a new position.

(5) **Deleting a file :** Search the file and then delete it.

(6) **Truncating a file :** This operation erase all the contents of file but it keeps its attributes.

Types of Files:

File Type	Extension	Function.
(1) executable	.exe, com, bin	mechine language program
(2) object	obj, o	compited mechine language not linked
(3) source code	c, cc, java, asm	source code in various languages
(4) batch	bat, sh	commands to the command interpreter.
(5) text	txt, doc	Textual data, documents
(6) library	lib, dll	libraries of routine for programmers.
(7) archive.	arc, zip, tar	related files grouped into one file. compressed file.
(8) multimedia	mpeg, mov, mp3	Binary file containing audio or video information.

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File Access Mechanisms :

File access mechanism refers to the manner in which the records of a file may be accessed. There are several ways to access files.

1. Sequential access.
2. Direct / Random access.
3. Indexed sequential access.

(1) Sequential access :

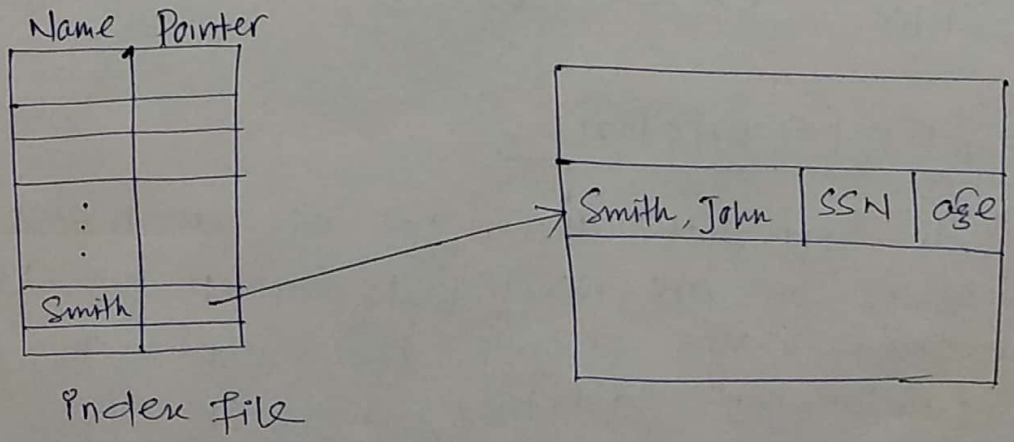
A sequential access mechanism accesses records in some sequence i.e the information in the file are processed in order, one record after the other. This method is very basic. (Example : Compiler accesses files in this fashion.)

(2) Direct Access :

In direct access, the file is viewed as a numbered sequence of blocks or records. A direct access file allows arbitrary blocks to be read or written.

(3) Indexed Access :

This method uses an index for the file. The index contains pointer to the various blocks. To find a record in the file we first search the index, and then use the pointer to access the file directly, and to find the desired record.



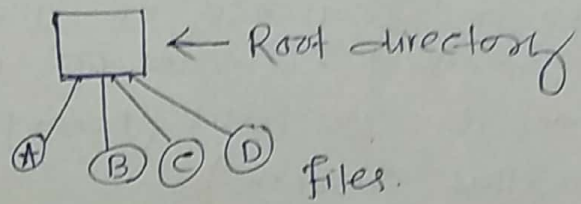
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Directory structure :

To keep track of files, file systems normally have directories or folders. A directory records information such as name, location, size and type for all files that it contains.

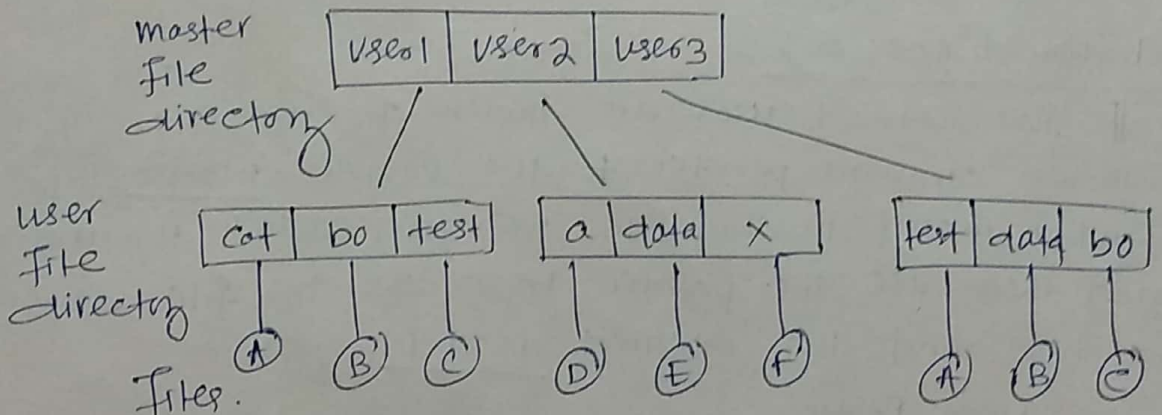
(1) Single level directory :

The simplest directory structure is the single level directory. All files are contained in the same directory.



(2) Two level directory :

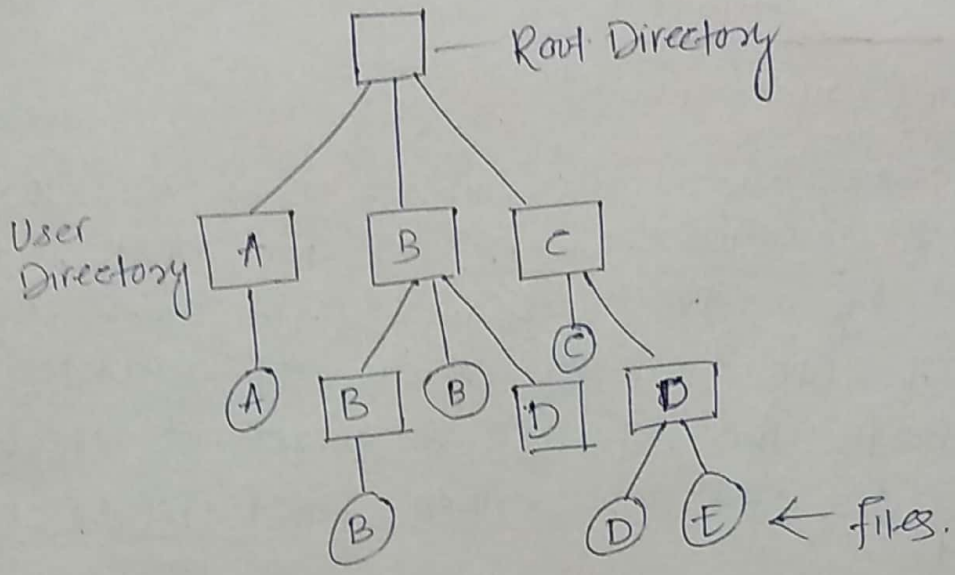
A single level directory often create confusion of file name between different users. In two level directory structure each user has own user file directory.



(3) Hierarchical directory :

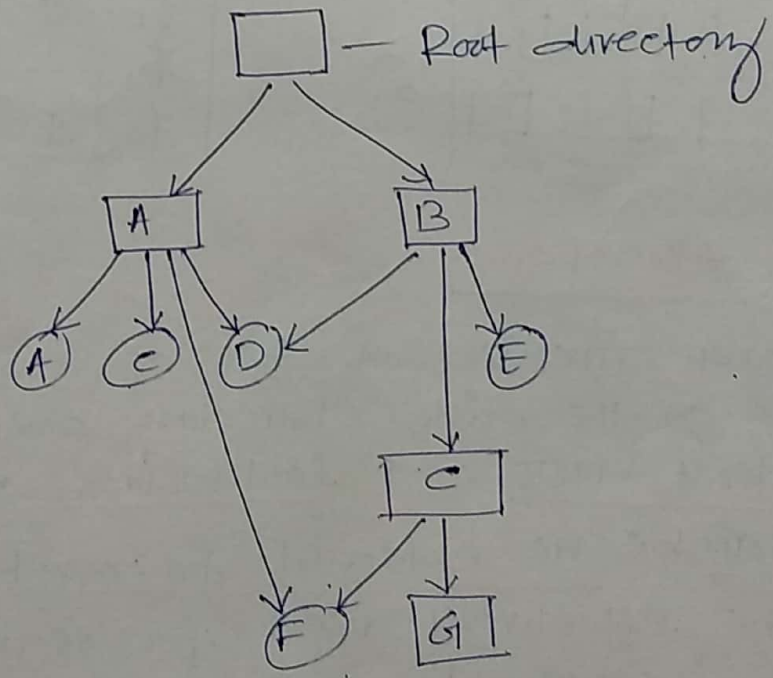
In this approach there can be as many directories as are needed to group the files in natural ways. Using this approach user can create arbitrary number of sub-directories. All marden file system follows this approach.

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(4) Acyclic graph directory :

This approach allows directories to have shared sub directories and files. The same file or sub directory may be in two different directories. This is implimented by maintaining links to subdirectories and files.



A cyclic graph directory structure.

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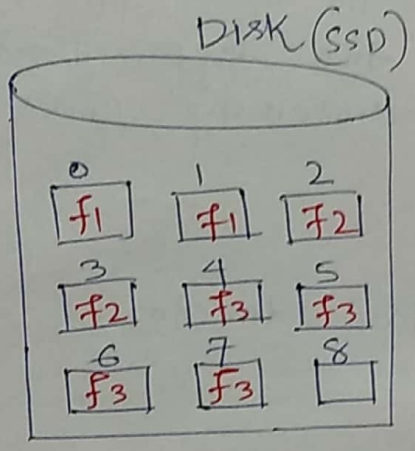
File Allocation Methods :

(1) Contiguous Allocation :

Contiguous allocation requires each file to occupy a set of contiguous blocks of the disk. It is defined by address of the first block and size of file in terms of no of blocks.

To satisfy the storage requirement of n free blocks os can use either first fit or best fit method.

This method suffers from ~~internal~~ external fragmentation as well as some inter fragmentation. It supports sequential as well as direct access.



Directory

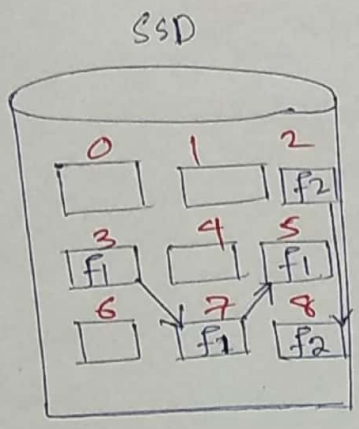
File Name	Start Add	Size
f1	0	2
f2	2	2
f3	4	3

(2) Linked Allocation :

Here each file is link list of disk blocks scattered on the disk. Directory contains pointers to the first block and last block of the file.

There will be no external fragmentation. It can be effectively used for sequential access but not directly accessed.

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Directory

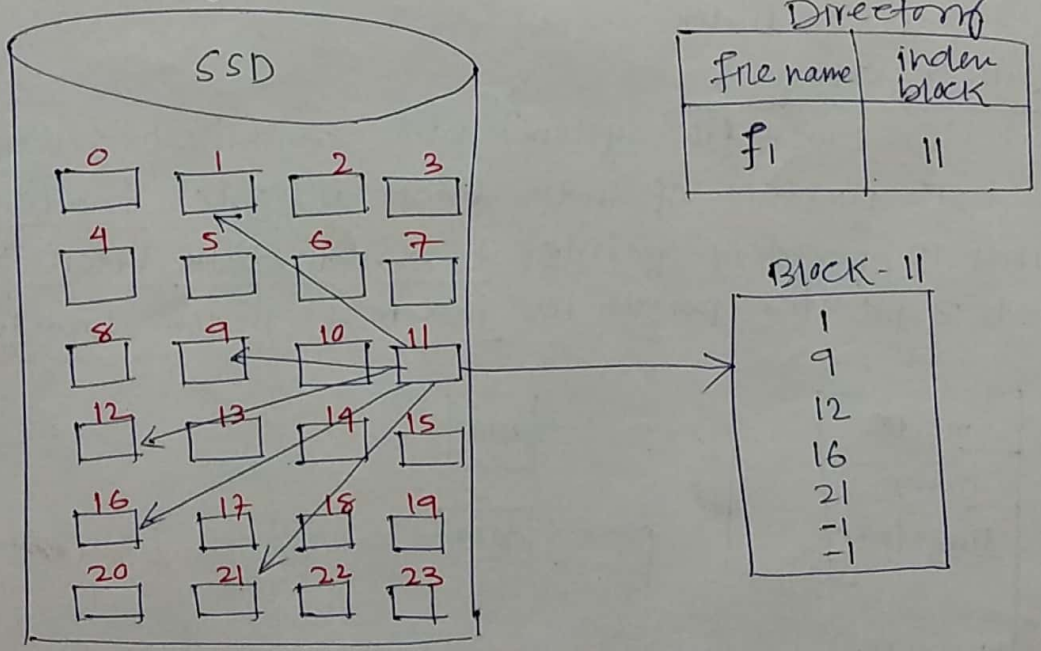
File Name	Start Addr	End Addr
f_1	3	5
f_2	2	8

(3) Indexed Allocation :

This method brings all the pointer together into one location for index block. Each file have its own index block. The directory contains name of the file and address of its index block.

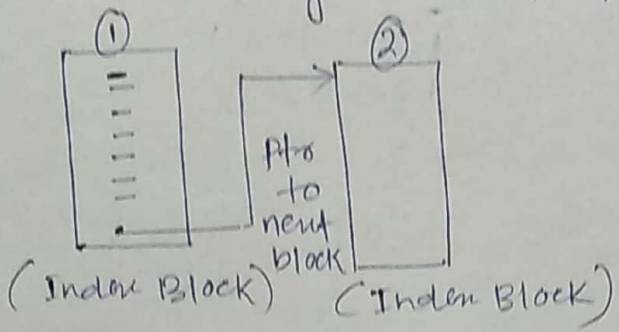
There will be no external fragmentation. It supports indexed access which is as good as direct access. Unix OS uses this allocation.

To accomodate a large file several indexed block may be linked together.

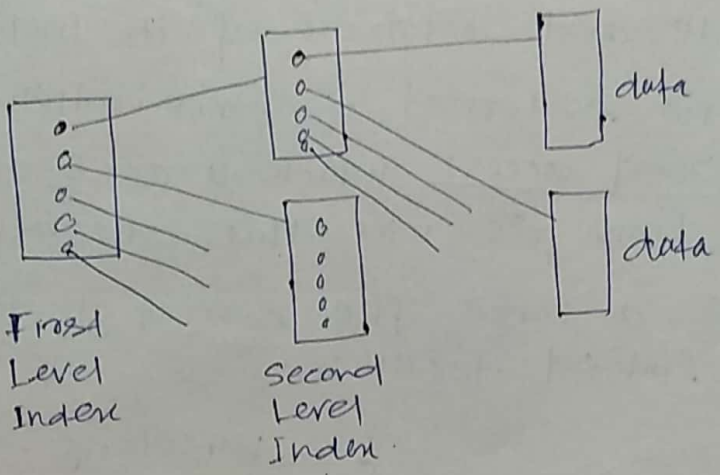


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(a) Linked scheme: To accomodate a large file several index block may be linked together.

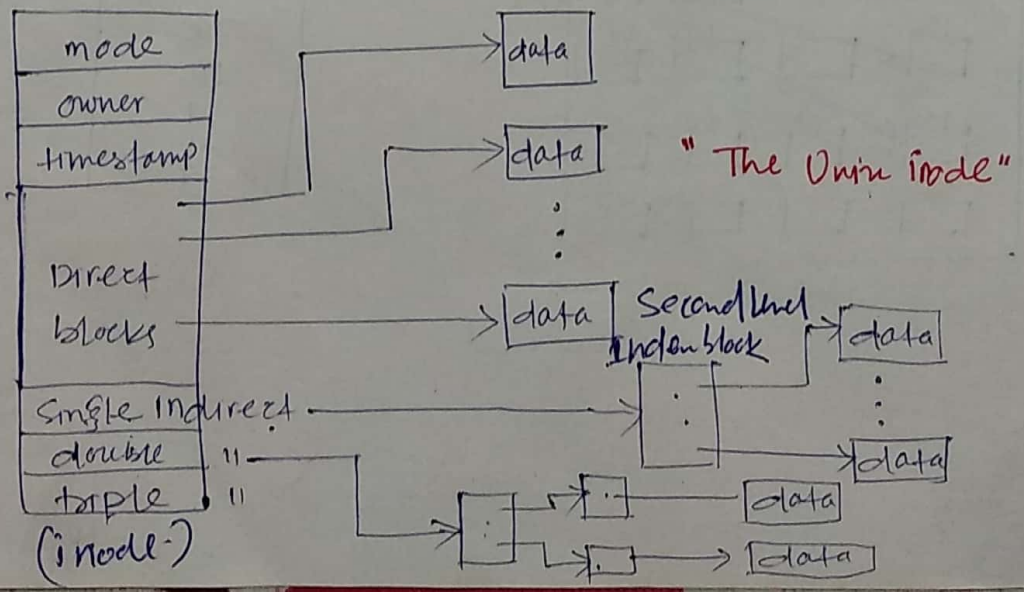


(b) Multilevel indexing scheme: In this scheme there are several level of indexing is used. First layer index will store the address of second layer index and second layer index will contain actual data file block.



(c) Combined scheme:

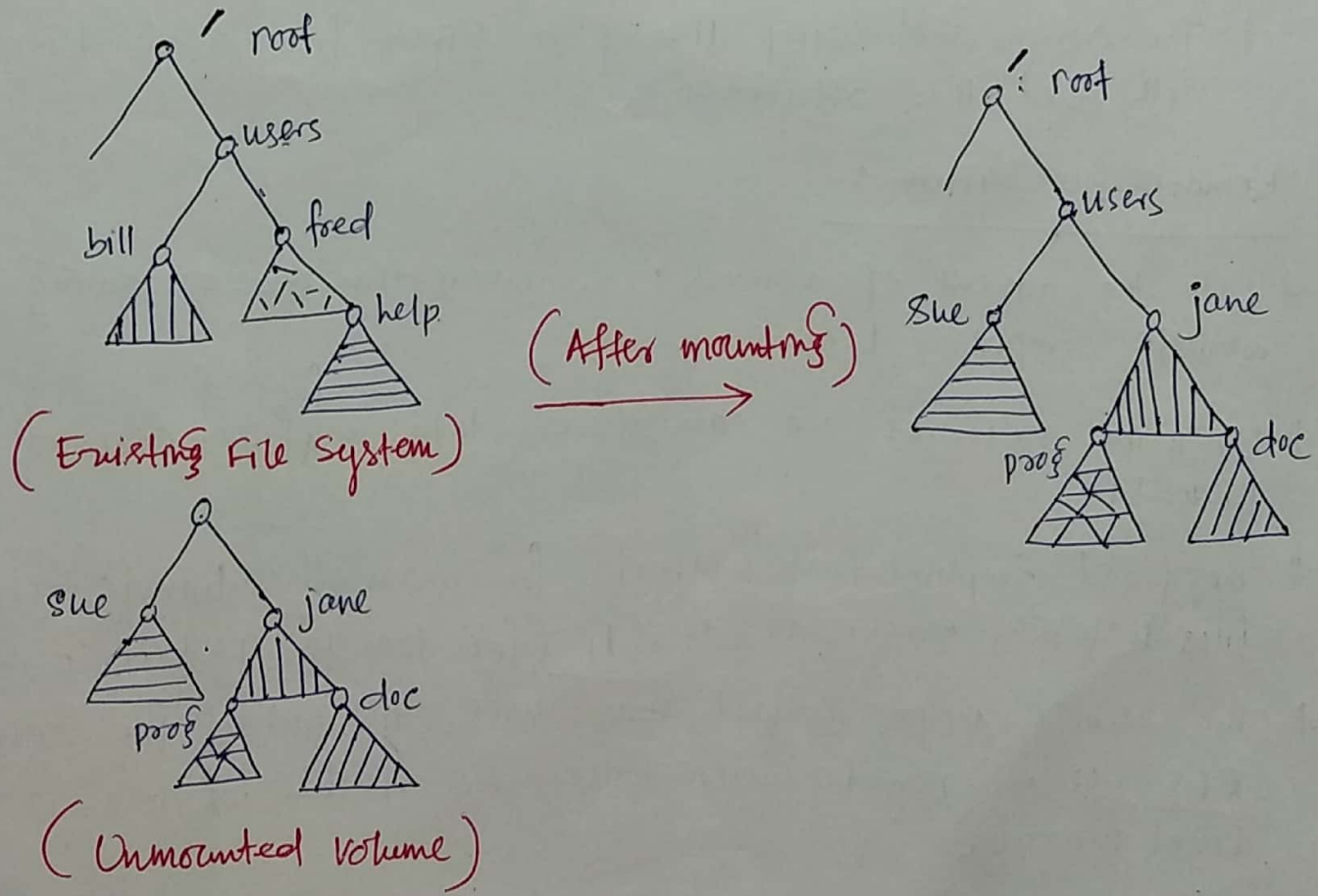
The UFS (Unix file system) uses combined scheme. First 15 pointers of index block is files inode. First 12 pointers points to direct data block and next 3 pointer points to indirect index block.



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File System Mounting:

- A file system must be mounted before it can be available to processes in the system.
- In mounting, operating system gives the name of the device and mount point. The mount point is the location where the file system is to be attached.
- Typically, a mount point is an empty directory.
- Next the OS verifies that the device contains a valid file system by asking the device driver to read the device directory structure and verify the expected format.
- Finally, OS notes in its directory structure that a file system is mounted at the specified mount point.



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File Sharing :

File sharing is very desirable for users who want to collaborate and to reduce the effort required to achive a computing goal.

The following are different aspects of file sharing.

Multiple users :

In multiple user file system, OS must maintain more file and directory attribute to impliment sharing and protection.

Most system use the concept of owner, group and group where owner is the user who can change attributes and grant access and has most control over file.

The group attribute defines a subset of users who can share access to file.

The owner and group IDs of a given file are stored with the other attributes.

Remote file System :

With the advent of networks, communication among remote computers became possible.

By using networks we can share data in form of files around the world.

The first implimented method for manually transfering files between machines was ftp (file transfer protocol) program.

The second major method uses distributed file system (DFS) where remote directories are visible from local machine.

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The client server model :

- Remote file systems allow a computer to mount one or more file system from one or more remote place.
- In this case, the machine containing the files is the "server" and the machine asking to files is the "client".
- A server can serve multiple clients and a client can use multiple servers depending on the implimentation details of a given client server facility
- The server specifies the available files on a volume or directory.
- UNIX operating system uses network file system (NFS). In NFS server first authenticate the users of file system by using clients networking information like IP address.

Distributed Information Systems :

- To make client-server systems easier to manage, distributed information systems, provide unified access to the information needed for remote computing.
- The domain name system (DNS) provides host-name-to-network address translation for entire Internet.
- SUN microsystem introduced a distributed information system called "yello pages" which centralizes storage of user names, host names, printer information etc.
- The distributed information system used by Microsoft is common Internet file System (CIFS).

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Protection:

When information is stored in a computer, we want to keep it safe from physical damage and improper access.

Physical damage can be handled by reliable mechanism in which we use duplicate copies of files.

Many computers have system programs that automatically copy disk files to backup devices at regular interval to maintain a copy. This can prevent loss of information due to accidental destroy.

To handle improper access we use protection mechanism for single user systems like laptop we can provide protection by locking the computer in desk drawer or file cabinet.

In larger multi user system other mechanisms are needed.

Types of Access:

The need to protect files is direct result of the ability to access files. So we need control access to the files.

Protection mechanism provide controlled access by limiting the type of file access that can be made.

Several different types of operation may be controlled:

Read: Read from the file
Write: Write or rewrite the file
Execute: Load the files into memory and execute it.
Append: Write new information at the end of the file.
Delete: Delete the file and free its space for reuse.
List: List the name and attributes of the file.

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Access Control :

The most common approach to protection problem is to make access dependent on the identity of the user.

To impliment identity dependent access is to associate each file ~~with each file~~ and directory an access control list (ACL) specfing user name and type of access allowed for each user.

The ACL ~~may~~ in many system recognize three classification of users in connection to each file.

Owner : The user who created the file is the owner.

Group : A set of user who are sharing the file and need similar access is a group, or work group.

Universe : All other user in the system constitute the universe.

To impliment this scheme, three fields are needed to define protection.

Each field is a collection of bits, and each bit either allows or prevents the access associated with it.

for example, the UNIX system defines three fields of 3 bit each : rwx where 'r' controls read access, "w" controls write access and "x" controls execution.

In this scheme, nine bits perfile are needed to record protection information.

Example : book.txt | rwx rw- r-- | sara tent 130 may 18 ...

000 000 000

↑ filename

Owner Group Universe

Chmod Command

for the owner "sara", all bits are set, for the group "tent" the r and w bits are set and for the universe only r bit is set (i.e universe can perform read operation only).

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